Project Proposal

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Rules

- **3 students per team**; total 15 teams
- **Oral presentation**:10 minutes per team
  - 8-minute presentation + 2-minute QA
  - Either English or Chinese is okay
  - If we have fewer teams maybe longer time
- Submit slides after presentation

- **Registration before 4/2 00:00**
- https://forms.gle/ffj6AxNn1zL19K1s8

列點其他ML或相關課程做過的專題題目和資料集連結 *

詳答文字

1. Advertisement click prediction. https://...
2. Social recommendation. https://...

Scores

- 10% of total score

- In this proposal presentation:
  - Scenario (10%)
  - Problem definition (15%)
  - Dataset observation (40%)
  - Challenges (25%)
  - To-dos (10%)

Scenario and Problem Definition

- **Scenario:** design a simple story to introduce your problem
  - The goal is to let audience understand your task quickly

We are the agents of advertisers. We need to allocate the budget of buying impressions. Eventually, we want to maximize the number of clicks.

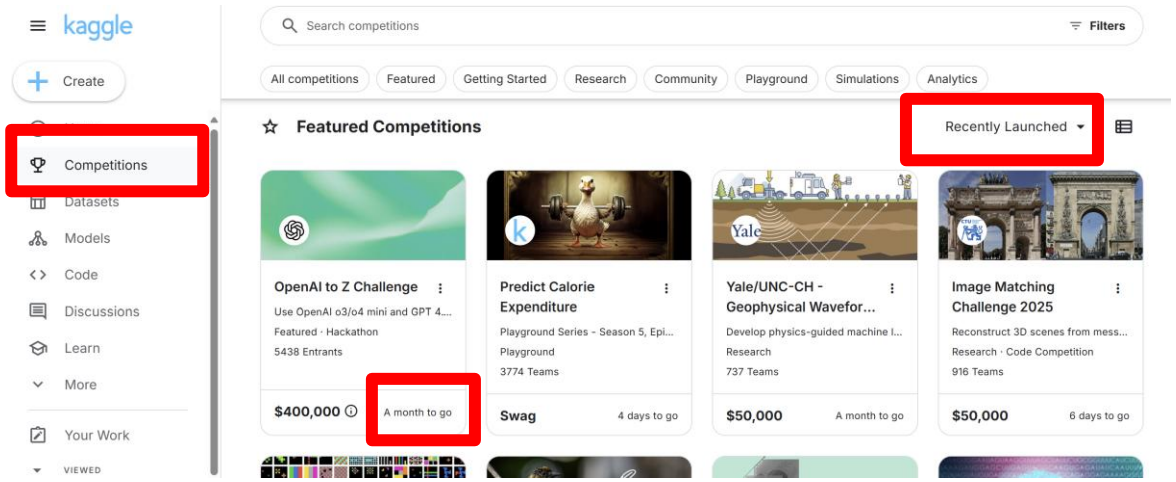
- **Problem:**
  - Your task must **"predict"** something
  - Mostly supervised learning
  - Find problems on Kaggle, github, or open source codes
  - **Novelty** is valued!
  - Must **NOT** do identical problems in ML projects

Given the dataset of XXX, our goal is to forecast whether an advertisement will be clicked.

Dataset Requirements

- Find your dataset on Kaggle or other open source platforms
- Contain at least 50,000 instances

- Perform **data analysis**
  - Basic statistics
  - Properties
  - Patterns
  - Interesting findings



Challenges

- **Based on your statistics**, what properties cause the prediction task **extra difficulties**?
  - Data imbalance
  - Class skewness
  - High dimensional feature space
  - …
- Which finding?
- Define difficulties and why
  - Try to list 2-3 challenges

Clicked:non-clicked=1:99
The classifier always predicts the non-clicked because it causes at least 99% accuracy.

To-dos

- What skills do you expect to use to solve the challenges?
  - List 2-3 for each challenge
  - Not-yet-implemented is acceptable
  - Sources: course slides, Internet, and AI

Stratified sampling, negative sampling, ranking-based objectives, … may solve the class imbalance issue

Practice Your Presentation

- 8 minute is not much
- Make sure you invest the limited time to the right place

- If you have too many materials that cannot fit in 8-minute presentation, put
  - main information in the presentation
  - the remainings in an appendix

- Present to someone else and improved based on their feedback